

§1.1

The Base Ten Place Value System

Chapter 1 — Numeration Systems

Today's Goal

After completing this section, you should be able to:

- Express a number using the base ten place value system in standard form, expanded form, and word form
- Use appropriate models such as base ten blocks to represent a number in the base ten place value system

Learning Target 1

LT.1 Base Ten Place Value System

I can use the base ten place value system to represent numbers in standard, expanded, and word form as well as use appropriate models such as base ten blocks to represent numbers.

Everything in §1.1 is practice for this one target.

Every Digit Is a Five

In 5,555, does each 5 mean the same thing?

No. Read left to right, each five is worth ten times the one after it:

$$5,555 = 5,000 + 500 + 50 + 5$$

The digit did not change. Its **position** did. That idea is the whole section.

Every Position Is a Power of Ten

In the base ten system, also called the **decimal system**, each place represents a power of ten.

Position	n	...	4th	3rd	2nd	1st
Place Value	10^{n-1}	...	$10^3 = 1,000$	$10^2 = 100$	10	1

Positions are counted from the right. The 1st position is the ones place.

Why Place Value Carries So Much

You will teach this to children who are meeting it for the first time. Here is what rides on it.

- **Arithmetic.** Adding means lining digits up by place.
- **Number sense.** Judging size and estimating both come from place value.
- **Reasoning.** $500 + 30 + 4 = 534$ — taking a number apart and rebuilding it.
- **Algebra later.** The same structure returns as coefficients.

Writing a Number in Standard Form

Procedure 1.1.2. Writing Whole Numbers in Standard Form

To write a whole number in standard form, do the following.

1. Identify the place value(s) needed in the number.
2. Identify how many of each place value is needed in the number.
3. Write the digits in their respective positions, starting from the leftmost digit (the highest place value) to the rightmost digit (the lowest place value).

Where the Zero Comes From

Write **six thousand ninety-three** in standard form.

1. Place values needed: thousands, hundreds, tens, and ones.
2. How many of each: six thousands, *no* hundreds, nine tens, and three ones.
3. Write the digits in position: 6,093.

Nothing was said about hundreds, so the hundreds place holds a 0. Without it the number would read 693.

Now You Try: Standard Form

You Try 1.1.1. Write the number four thousand seven hundred twenty-six in standard form.

Answer. 4,726 — four thousands, seven hundreds, two tens, six ones.

What Is That Five Worth?

In 8,517, the third digit is a 5. What is it worth?

500 — because it sits in the hundreds place, so it means 5×100 .

Writing out every digit that way gives us the **expanded form**.

Expanded Form, Defined

Definition 1.1.4

A whole number, N , in the base-10 number system, is written in **expanded form** when it is written in the form

$$N = d_n \cdot 10^{n-1} + d_{n-1} \cdot 10^{n-2} + \cdots + d_3 \cdot 10^2 + d_2 \cdot 10 + d_1$$

where $d_1, d_2, \dots, d_n$ are the digits in the number N (read from right to left) and n is the number of digits in N .

Writing a Number in Expanded Form

Procedure 1.1.5. Writing Whole Numbers in Expanded Form

To write a whole number in expanded form, do the following.

1. Identify the place value of each digit in the number.
2. Multiply each digit by its corresponding place value.
3. Write the number as a sum.

One Digit at a Time

Write 8,517 in expanded form. Each digit is worth its *face value* times its *place value*.

Position	Place Value	Digit	Value of Digit
1st	1	7	$7 \times 1 = 7$
2nd	$10^1 = 10$	1	$1 \times 10 = 10$
3rd	$10^2 = 100$	5	$5 \times 100 = 500$
4th	$10^3 = 1,000$	8	$8 \times 1,000 = 8,000$

Three Ways to Write the Same Thing

The last column of the table *is* the expanded form. All three of these say the same thing.

$$8,517 = 8 \cdot 1,000 + 5 \cdot 100 + 1 \cdot 10 + 7$$

$$8,517 = 8 \cdot 10^3 + 5 \cdot 10^2 + 1 \cdot 10 + 7$$

$$8,517 = 8,000 + 500 + 10 + 7$$

Now You Try: Expanded Form

You Try 1.1.2. Complete the following to write 27,431 in expanded form.

Position	Place Value	Digit	Value of Digit
1st	1	1	1
2nd	10	3	30
3rd	100	4	400
4th	1,000	7	7,000
5th	10,000	2	20,000

Now You Try: Three Ways

You Try 1.1.2, continued. Now write 27,431 in expanded form three different ways.

$$27,431 = 2 \cdot 10,000 + 7 \cdot 1,000 + 4 \cdot 100 + 3 \cdot 10 + 1$$

$$27,431 = 2 \cdot 10^4 + 7 \cdot 10^3 + 4 \cdot 10^2 + 3 \cdot 10 + 1$$

$$27,431 = 20,000 + 7,000 + 400 + 30 + 1$$

The Names You Have to Know

Word form writes a number in words instead of digits. These names are single words.

Number	Word Form	Number	Word Form	Number	Word Form
1	one	8	eight	15	fifteen
2	two	9	nine	16	sixteen
3	three	10	ten	17	seventeen
4	four	11	eleven	18	eighteen
5	five	12	twelve	19	nineteen
6	six	13	thirteen		
7	seven	14	fourteen		

And the Tens

Number	Word Form	Number	Word Form
20	twenty	60	sixty
30	thirty	70	seventy
40	forty	80	eighty
50	fifty	90	ninety

Those two tables are the whole vocabulary. Every other number name is built out of them.

Building Every Other Name

- **21 to 99 are hyphenated.** $49 = 40 + 9$, so 49 is “forty-nine.”
- **Multiples of 100 take two words.** 400 is “four hundred.”
- **Others split at the hundred.** $349 = 300 + 49$, so 349 is “three hundred forty-nine.”

Expanded form tells you how to say it. Break the number up, name each piece, then join the pieces.

Never Say “and”

Watch this one

Write “three hundred forty-nine,” not “three hundred *and* forty-nine.”

The word “and” has a specific job that arrives with fractions and decimals. Spending it here leaves nothing to say later.

Digits Come in Groups of Three

For numbers with more than three digits, split into groups of three from the right. Each group is a **period**, and each period is read as a three-digit number followed by the period's name.

— — — , — — — , — — — , — — — , — — — , — — —
quadrillions trillions billions millions thousands ones

Later periods are called quintillion, sextillion, septillion, and so on. You will rarely need them.

Writing a Number in Word Form

Procedure 1.1.10. Writing Whole Numbers in Word Form

To write a whole number in word form, do the following.

1. Identify the periods present in the number.
2. Write the three-digit number for each period in word form. Any periods in which all digits are 0 may be skipped in this step.
3. Combine the word form for each period from Step 2.

One Period at a Time

Write each number in word form.

(a) 72,509

- Thousands period: 72 — “seventy-two thousand”
- Ones period: 509 — “five hundred nine”

seventy-two thousand, five hundred nine

Three Periods

(b) 615,304,278

- Millions period: 615 — “six hundred fifteen million”
- Thousands period: 304 — “three hundred four thousand”
- Ones period: 278 — “two hundred seventy-eight”

six hundred fifteen million, three hundred four thousand, two hundred seventy-eight

A Period You Skip

(c) 43,700,000,065

- Billions period: 43 — “forty-three billion”
- Millions period: 700 — “seven hundred million”
- Thousands period: every digit is 0, so **skip it**
- Ones period: 065 — “sixty-five”

forty-three billion, seven hundred million, sixty-five

The zeros still hold the places. They just get no name.

Now You Try: Word Form

You Try 1.1.3. Write each of the following numbers in word form.

1. 483,061 **four hundred eighty-three thousand, sixty-one**
2. 250,974,308 **two hundred fifty million, nine hundred seventy-four thousand, three hundred eight**

Now Every Place Has a Name

Place Value	Name
1	ones
10	tens
$10^2 = 100$	hundreds
$10^3 = 1,000$	one thousands
$10^4 = 10,000$	ten thousands
$10^5 = 100,000$	hundred thousands
$10^6 = 1,000,000$	one millions

Show It Without Writing It

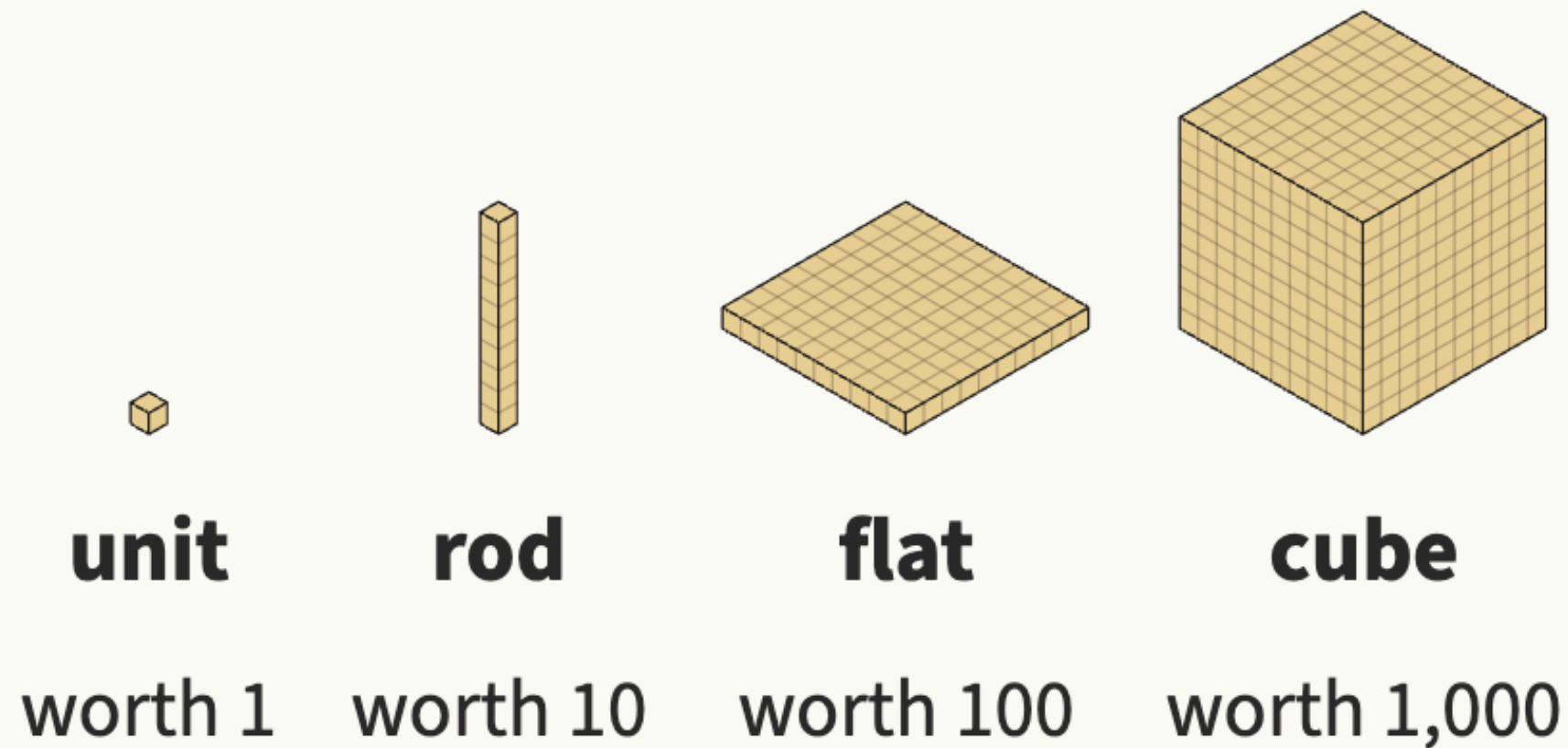
A first-grader cannot read 243 yet. How do you *show* her that number?

You hand her something she can hold and count. That is what a **model** is for.

We will look at two: base ten blocks, and dots and boxes.

Base Ten Blocks

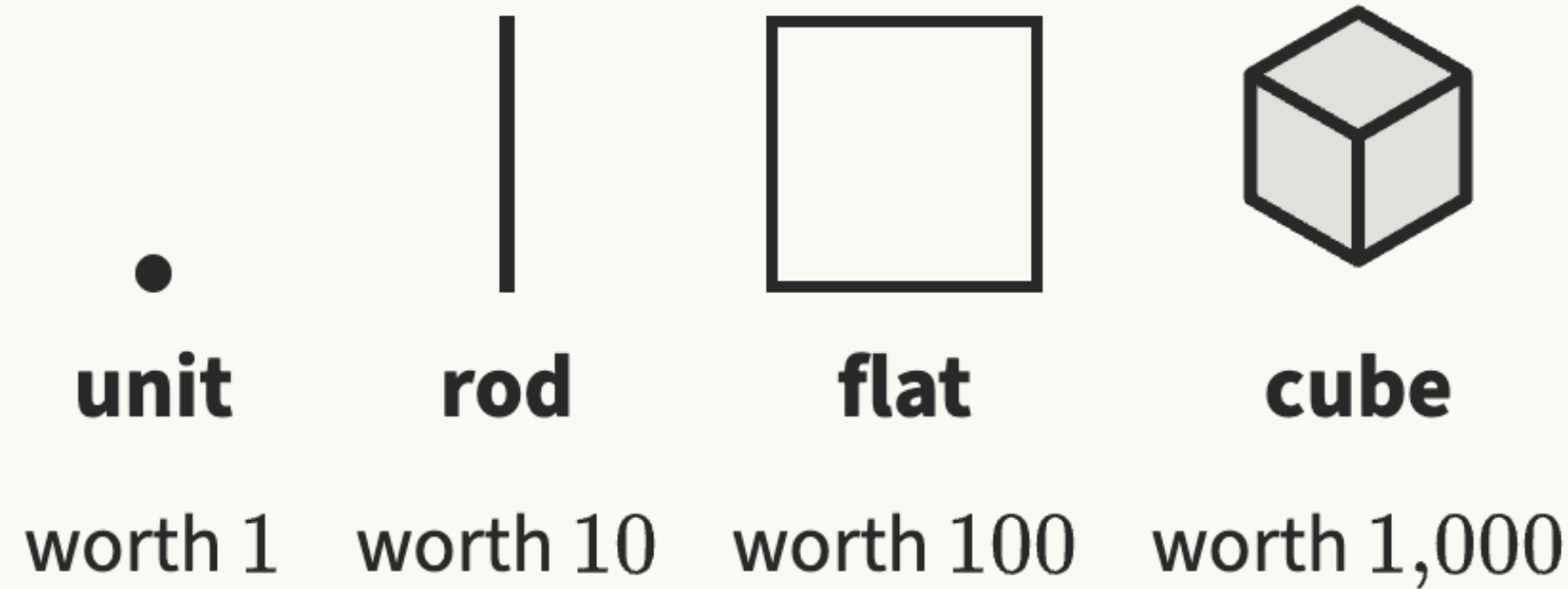
Four pieces, each made of the one before it — ten of any piece build the next.



So 1,234 is 1 cube, 2 flats, 3 rods, and 4 units.

Drawing Blocks by Hand

You will not always have blocks. Draw them instead — this is the shorthand the notes use.



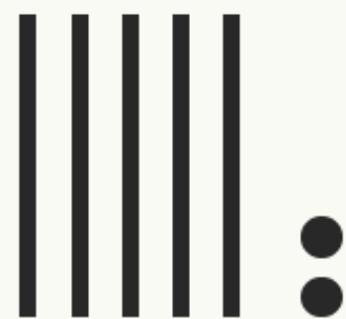
A dot, a line, a square, a cube. That is the whole notation.

Building a Number from Blocks

Represent each number using the fewest base ten blocks.

(a) 52

Since $52 = 50 + 2$, we need 5 rods and 2 units.



A Place with Nothing in It

(b) 306

Since $306 = 300 + 6$, we need 3 flats, *no* rods, and 6 units.

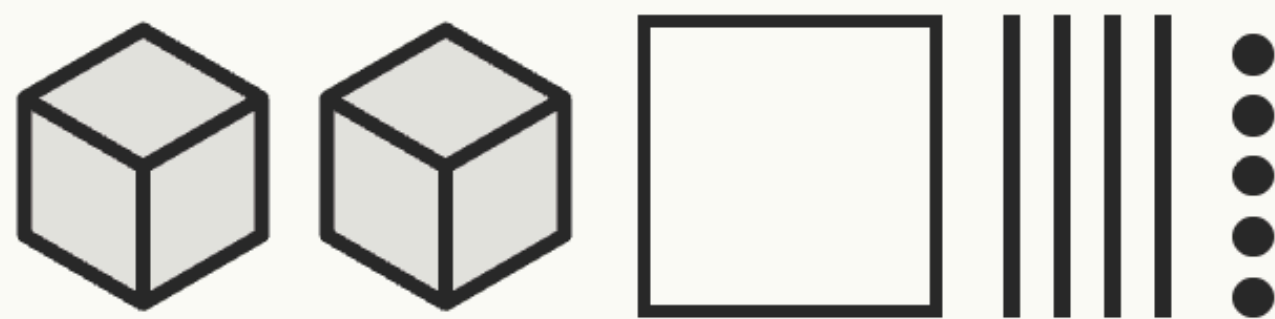


No rods are drawn. In the numeral, that empty tens place is the 0.

Adding the Thousands

(c) 2,145

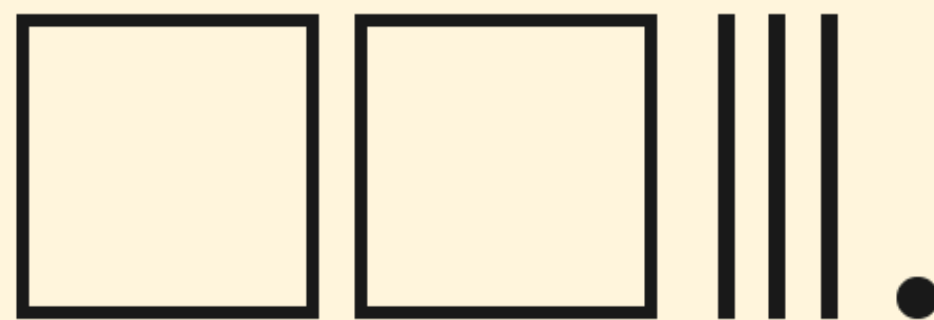
Since $2,145 = 2,000 + 100 + 40 + 5$, we need 2 cubes, 1 flat, 4 rods, and 5 units.



The digits of the number and the counts of the blocks are the same four numbers, in the same order.

Now You Try: Base Ten Blocks

You Try 1.1.4. Represent 573 using base ten blocks. The blocks below are a start, but some of them are missing.



How many more of each block are needed?

Answer. 573 needs 5 flats, 7 rods, and 3 units — so 3 more flats, 4 more rods, and 2 more units.

Dots and Boxes

In the **dots and boxes** model, each place value gets a box and each digit is a count of dots. Put the right number of dots in each box and you have written the number.

ten thousands	thousands	hundreds	tens	ones

Boxes keep going to the left for larger place values.

From a Number to the Model

Represent 62,308 using dots and boxes. Start with the expanded form — it tells you how many dots each box needs.

$$62,308 = 6 \cdot 10,000 + 2 \cdot 1,000 + 3 \cdot 100 + 0 \cdot 10 + 8$$

ten thousands	thousands	hundreds	tens	ones
				

From the Model to a Number

What number is represented below?

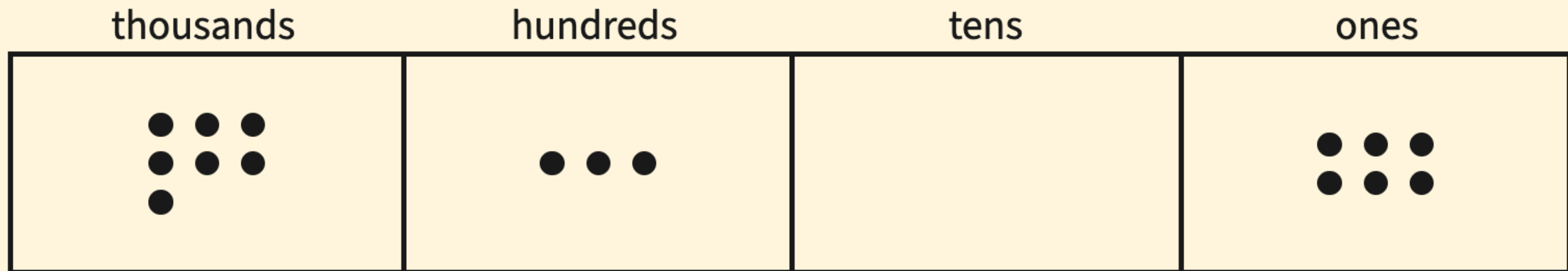
ten thousands	thousands	hundreds	tens	ones
				

Count the dots in each box, then write the expanded form:

$$4 \cdot 10,000 + 0 \cdot 1,000 + 7 \cdot 100 + 9 \cdot 10 + 3 = 40,793$$

Now You Try: Dots and Boxes

You Try 1.1.5. What number is being represented by the base ten dots and boxes model shown below?



Answer. $7 \cdot 1,000 + 3 \cdot 100 + 0 \cdot 10 + 6 = 7,306$

The System You Already Use

The base ten system has a name: the **Hindu-Arabic numeration system**. It began in India around the 6th century, was developed further in the Islamic world, and reached Europe by the 12th century.

- **Ten digits.** 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 — and nothing else.
- **Place value.** A digit's value depends on its position, and positions are powers of ten.
- **Zero.** Both a digit and a placeholder. This is the part other systems did not have.
- **Decimal notation.** A decimal point separates whole from fractional parts.

Why This One Won

- **Simple.** Ten digits and position — that is the entire rule set.
- **Efficient.** Very large and very small numbers fit in a short string.
- **Universal.** The same numerals are read across languages and cultures.
- **Versatile.** The same place-value rule keeps running past the ones place. Step right of the decimal point and you get tenths and hundredths — so money, measurement, and parts of a whole all live in one system.

In §1.2 through §1.5 we look at the systems it replaced — and you will see exactly what they were missing.